Chapter 4. The Fourier Transform

The class $\mathfrak{F}$, Fourier inversion and Poisson summation, the Paley–Wiener theorem

Contents

1 The class $\mathfrak{F}$ **2**

2 Action of the Fourier transform on $\mathfrak{F}$ **2**

3 Paley–Wiener theorem **4**

4 Selected exercises **6**

Overview

For f on $\mathbb{R}$ with suitable regularity and decay, the Fourier transform and its inversion are

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x\xi}\, dx, \qquad f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

Holomorphic extension of f corresponds to very rapid decay of $\hat{f}$. If f continues to a horizontal strip with moderate decrease, $\hat{f}$ decays exponentially, and inversion and Poisson summation follow by contour integration. The *Paley–Wiener theorem*: $\hat{f}$ is supported in $[-M, M]$ iff f extends to an entire function with $|f(z)| \le Ae^{2\pi M|z|}$ (*exponential type*). The technique is shifting the real line within a strip, where $e^{-2\pi i z\xi}$ decays or grows according to the sign of $\xi \operatorname{Im}(z)$.

A function f is of **moderate decrease** if it is continuous and $|f(x)| \le A/(1+x^2)$ for some $A > 0$.

1 The class $\mathfrak{F}$

Book I proved inversion and Poisson summation for $|f(x)| \le \frac{A}{1+x^2}$, $|\hat{f}(\xi)| \le \frac{A'}{1+\xi^2}$ (e.g. $P_y(x) = \frac{1}{\pi}\frac{y}{y^2+x^2}$, $\hat{P}_y(\xi) = e^{-2\pi y|\xi|}$). Here is a class suited to complex analysis.

Definition (the classes $\mathfrak{F}_a$ and $\mathfrak{F}$)

For $a > 0$, $\mathfrak{F}_a$ is the class of functions f such that:
(i) f is holomorphic in the horizontal strip $S_a = \{z \in \mathbb{C} : |\operatorname{Im}(z)| < a\}$;
(ii) there is $A > 0$ with $|f(x+iy)| \le \dfrac{A}{1+x^2}$ for all $x \in \mathbb{R}$ and $|y| < a$. $\mathfrak{F}$ is the class of functions belonging to $\mathfrak{F}_a$ for some a.

So $\mathfrak{F}_a$ consists of holomorphic functions on S_a of moderate decay on horizontal lines, uniformly in $-a < y < a$. Examples: $e^{-\pi z^2} \in \mathfrak{F}_a$ for all a; $\frac{1}{\pi}\frac{c}{c^2+z^2} \in \mathfrak{F}_a$ for $0 < a < c$; $1/\cosh \pi z \in \mathfrak{F}_a$ for $|a| < 1/2$. By the Cauchy integral formulas, $f \in \mathfrak{F}_a$ implies $f^{(n)} \in \mathfrak{F}_b$, $0 < b < a$. (Most results hold with $A/(1+|x|^{1+\epsilon})$.)

2 Action of the Fourier transform on $\mathfrak{F}$

Motivation. Contour shifting in a strip proves inversion and Poisson summation without real-variable convergence issues.

All three theorems are proved by contour integration.

Theorem 2.1

If f belongs to the class $\mathfrak{F}_a$ for some $a > 0$, then $|\hat{f}(\xi)| \le Be^{-2\pi b|\xi|}$ for any $0 \le b < a$.

Proof. $b = 0$ is clear from moderate decrease. Let $0 < b < a$, $\xi > 0$, $g(z) = f(z)e^{-2\pi i z\xi}$, and shift the real line down by b using the rectangle with vertices $-R, R, R-ib, -R-ib$. The left vertical side is bounded by

$$\Big|\int_{-R-ib}^{-R} g(z)\, dz\Big| \le \int_0^b |f(-R-it)e^{-2\pi i(-R-it)\xi}|\, dt \le \int_0^b \frac{A}{R^2} e^{-2\pi t\xi}\, dt = O(1/R^2),$$

and similarly for the right side. By Cauchy's theorem, letting $R \to \infty$,

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x-ib)e^{-2\pi i(x-ib)\xi}\, dx, \tag{1}$$

so $|\hat f(\xi)| \le \int_{-\infty}^{\infty} \frac{A}{1+x^2} e^{-2\pi b\xi}\,dx \le Be^{-2\pi b\xi}$. For $\xi < 0$ shift the real line up by b. ■

The further f extends (larger a), the better the decay of $\hat f$.

Theorem 2.2

If $f \in \mathfrak{F}$, then the Fourier inversion holds, namely

$$f(x) = \int_{-\infty}^{\infty} \hat f(\xi) e^{2\pi i x\xi}\,d\xi \qquad \text{for all } x \in \mathbb{R}.$$

Lemma 2.3

If A is positive and B is real, then $\int_0^\infty e^{-(A+iB)\xi}\,d\xi = \dfrac{1}{A+iB}$.

Proof. $|e^{-(A+iB)\xi}| = e^{-A\xi}$, so the integral converges, and $\int_0^R e^{-(A+iB)\xi}\,d\xi = \left[-\frac{e^{-(A+iB)\xi}}{A+iB}\right]_0^R \to \frac{1}{A+iB}$. ■

Proof of Theorem 2.2. Split $\int_{-\infty}^{\infty} \hat f(\xi) e^{2\pi i x\xi} d\xi = \int_{-\infty}^{0} + \int_0^{\infty}$. Say $f \in \mathfrak{F}_a$, $0 < b < a$. By (1) and the lemma (the integrals converge absolutely),

$$\begin{aligned}\int_0^\infty \hat f(\xi) e^{2\pi i x\xi}\,d\xi &= \int_0^\infty \int_{-\infty}^\infty f(u-ib) e^{-2\pi i(u-ib)\xi} e^{2\pi i x\xi}\,du\,d\xi = \int_{-\infty}^\infty f(u-ib) \int_0^\infty e^{-2\pi i(u-ib-x)\xi}\,d\xi\,du \\ &= \int_{-\infty}^\infty f(u-ib)\frac{1}{2\pi b + 2\pi i(u-x)}\,du = \frac{1}{2\pi i}\int_{-\infty}^\infty \frac{f(u-ib)}{u-ib-x}\,du = \frac{1}{2\pi i}\int_{L_1} \frac{f(\zeta)}{\zeta - x}\,d\zeta,\end{aligned}$$

$L_1 = \{u - ib : u \in \mathbb{R}\}$, left to right. Similarly $\int_{-\infty}^0 \hat f(\xi) e^{2\pi i x\xi} d\xi = -\frac{1}{2\pi i}\int_{L_2} \frac{f(\zeta)}{\zeta - x}\,d\zeta$, L_2 the line shifted up by b. For $x \in \mathbb{R}$ let γ_R be the rectangle with vertices $-R \pm ib$, $R \pm ib$; the residue of $f(\zeta)/(\zeta - x)$ at x is $f(x)$, so $f(x) = \frac{1}{2\pi i}\int_{\gamma_R} \frac{f(\zeta)}{\zeta - x}\,d\zeta$. As $R \to \infty$ the vertical sides vanish, and

$$f(x) = \frac{1}{2\pi i}\int_{L_1} \frac{f(\zeta)}{\zeta - x}\,d\zeta - \frac{1}{2\pi i}\int_{L_2} \frac{f(\zeta)}{\zeta - x}\,d\zeta = \int_0^\infty \hat f(\xi) e^{2\pi i x\xi}\,d\xi + \int_{-\infty}^0 \hat f(\xi) e^{2\pi i x\xi}\,d\xi.$$

■

Theorem 2.4 (Poisson summation formula)

If $f \in \mathfrak{F}$, then $\displaystyle\sum_{n\in\mathbb{Z}} f(n) = \sum_{n\in\mathbb{Z}} \hat f(n)$.

Proof. Let $f \in \mathfrak{F}_a$, $0 < b < a$. $f(z)/(e^{2\pi i z} - 1)$ has simple poles at the integers n with residues $f(n)/2\pi i$. The residue formula on the rectangle γ_N with vertices $\pm(N + \frac12) \pm ib$ gives

$$\sum_{|n|\le N} f(n) = \int_{\gamma_N} \frac{f(z)}{e^{2\pi i z} - 1}\,dz.$$

Let $N \to \infty$; by moderate decrease the vertical sides contribute 0, and

$$\sum_{n\in\mathbb{Z}} f(n) = \int_{L_1} \frac{f(z)}{e^{2\pi i z} - 1}\,dz - \int_{L_2} \frac{f(z)}{e^{2\pi i z} - 1}\,dz, \tag{2}$$

with L_1, L_2 the lines shifted down and up by b. On L_1, $|e^{2\pi i z}| > 1$ and $\frac{1}{e^{2\pi i z}-1} = e^{-2\pi i z}\sum_{n\ge 0} e^{-2\pi i n z}$; on L_2, $\frac{1}{e^{2\pi i z}-1} = -\sum_{n\ge 0} e^{2\pi i n z}$. Substituting in (2) and shifting back to the real line (by (1) and

its analogue),

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{n=0}^{\infty} \int_{L_1} f(z)e^{-2\pi i(n+1)z}\,dz + \sum_{n=0}^{\infty} \int_{L_2} f(z)e^{2\pi inz}\,dz = \sum_{n=0}^{\infty} \hat{f}(n+1) + \sum_{n=0}^{\infty} \hat{f}(-n) = \sum_{n\in\mathbb{Z}} \hat{f}(n).$$

■

Consequences. Since $e^{-\pi x^2}$ is its own Fourier transform (Example 1), for $t > 0$ and $a \in \mathbb{R}$ the change of variables $x \mapsto t^{1/2}(x + a)$ shows that $f(x) = e^{-\pi t(x+a)^2}$ has $\hat{f}(\xi) = t^{-1/2}e^{-\pi\xi^2/t}e^{2\pi ia\xi}$. Poisson summation gives

$$\sum_{n=-\infty}^{\infty} e^{-\pi t(n+a)^2} = \sum_{n=-\infty}^{\infty} t^{-1/2}e^{-\pi n^2/t}e^{2\pi ina}. \tag{3}$$

For $a = 0$ this is the transformation law of the theta function $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$ $(t > 0)$:

$$\vartheta(t) = t^{-1/2}\vartheta(1/t) \qquad \text{for } t > 0, \tag{4}$$

used for the functional equation of ζ (Chapter 6) and the Jacobi theta function (Chapter 10). Since $1/\cosh \pi x$ is its own Fourier transform (Example 3), $f(x) = e^{-2\pi iax}/\cosh(\pi x/t)$ has $\hat{f}(\xi) = t/\cosh(\pi(\xi + a)t)$, and

$$\sum_{n=-\infty}^{\infty} \frac{e^{-2\pi ian}}{\cosh(\pi n/t)} = \sum_{n=-\infty}^{\infty} \frac{t}{\cosh(\pi(n + a)t)}, \tag{5}$$

used in Chapter 10 for the two-squares theorem.

3 Paley–Wiener theorem

Motivation. Fourier transforms supported in a bounded interval are exactly those of entire functions of exponential type.

Now we assume no analyticity, but the validity of Fourier inversion when $|f(x)| \le A/(1 + x^2)$ and $|\hat{f}(\xi)| \le A'/(1 + \xi^2)$ (proved in Book I). A partial converse to Theorem 2.1:

Theorem 3.1

Suppose $\hat{f}$ satisfies the decay condition $|\hat{f}(\xi)| \le Ae^{-2\pi a|\xi|}$ for some constants $a, A > 0$. Then $f(x)$ is the restriction to $\mathbb{R}$ of a function $f(z)$ holomorphic in the strip $S_b = \{z \in \mathbb{C} : |\mathrm{Im}(z)| < b\}$, for any $0 < b < a$.

Proof. Let $f_n(z) = \int_{-n}^{n} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$, entire by Theorem 5.4, and $f(z) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$ on S_b; the integral converges absolutely, being majorized by $A\int e^{-2\pi a|\xi|}e^{2\pi b|\xi|}\,d\xi < \infty$ as $b < a$. On S_b, $|f(z) - f_n(z)| \le A\int_{|\xi|\ge n} e^{-2\pi a|\xi|}e^{2\pi b|\xi|}\,d\xi \to 0$, so f is holomorphic there by Theorem 5.2. ■

Corollary 3.2

If $\hat{f}(\xi) = O(e^{-2\pi a|\xi|})$ for some $a > 0$, and f vanishes in a non-empty open interval, then $f = 0$.

f is analytic in a region containing $\mathbb{R}$, so this follows from Theorem 4.8. In particular f and $\hat{f}$ cannot both have compact support unless $f = 0$.

Theorem 3.3 (Paley–Wiener)

Let f be continuous and of moderate decrease on $\mathbb{R}$. Then f extends to an entire function with $|f(z)| \le Ae^{2\pi M|z|}$ for some $A > 0$ if and only if $\hat{f}$ is supported in $[-M, M]$.

Proof. *($\Leftarrow$)* If $\hat{f}$ is supported in $[-M, M]$, both f and $\hat{f}$ have moderate decrease and $f(x) = \int_{-M}^{M} \hat{f}(\xi) e^{2\pi i \xi x}\, d\xi$. Define $g(z) = \int_{-M}^{M} \hat{f}(\xi) e^{2\pi i \xi z}\, d\xi$; $g = f$ on $\mathbb{R}$, g is entire by Theorem 5.4, and for $z = x + iy$, $|g(z)| \le \int_{-M}^{M} |\hat{f}(\xi)| e^{-2\pi \xi y}\, d\xi \le Ae^{2\pi M|z|}$.

($\Rightarrow$) For $\hat{f}$ supported in $[-M, M]$ the argument above gives the stronger bound $Ae^{2\pi M|y|}$. We first assume that bound plus decay in x, then remove the extra assumptions.

Step 1. Assume f is entire with

$$|f(x + iy)| \le A \frac{e^{2\pi M|y|}}{1 + x^2}. \tag{6}$$

Then $\hat{f}(\xi) = 0$ for $|\xi| > M$. For $\xi > M$, shifting the real line down by $y > 0$ (as in (1)),

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x - iy) e^{-2\pi i (x - iy)\xi}\, dx, \qquad |\hat{f}(\xi)| \le \int_{-\infty}^{\infty} A \frac{e^{2\pi M y - 2\pi y \xi}}{1 + x^2}\, dx \le Ce^{-2\pi y(\xi - M)},$$

and letting $y \to \infty$ gives $\hat{f}(\xi) = 0$. For $\xi < -M$ shift up.

Step 2. Assume only

$$|f(x + iy)| \le Ae^{2\pi M|y|}. \tag{7}$$

Let $\xi > M$ and $f_\epsilon(z) = \dfrac{f(z)}{(1 + i\epsilon z)^2}$. The factor has modulus ≤ 1 in the closed lower half-plane and tends to 1, so $|\hat{f}_\epsilon(\xi) - \hat{f}(\xi)| \le \int |f(x)| \left| \frac{1}{(1+i\epsilon x)^2} - 1 \right| dx \to 0$. Each f_ϵ satisfies $|f_\epsilon(x + iy)| \le A' \frac{e^{2\pi M|y|}}{1+x^2}$ in the lower half-plane, where Step 1 shifts the contour for $\xi > M$; so $\hat{f}_\epsilon(\xi) = 0$, hence $\hat{f}(\xi) = 0$. For $\xi < -M$ use the upper half-plane and $1/(1 - i\epsilon z)^2$.

Step 3. It remains to derive (7). It suffices: if $|f(x)| \le 1$ on $\mathbb{R}$ and $|f(z)| \le e^{2\pi M|z|}$, then $|f(x + iy)| \le e^{2\pi M|y|}$. In the first quadrant Q, $F(z) = f(z)e^{2\pi i M z}$ is bounded by 1 on both axes and $|F(z)| \le Ce^{c|z|}$, so Phragmén–Lindelöf (below, rotated) gives $|F| \le 1$ in Q, i.e. $|f(z)| \le e^{2\pi M y}$. The other quadrants are similar. ∎

The idea of Phragmén and Lindelöf adapts the maximum modulus principle to unbounded regions:

Theorem 3.4 (Phragmén–Lindelöf)

Let F be holomorphic in $S = \{z : -\pi/4 < \arg z < \pi/4\}$ and continuous on $\overline{S}$, with $|F| \le 1$ on ∂S and $|F(z)| \le Ce^{c|z|}$ in S for some $C, c > 0$. Then $|F(z)| \le 1$ for all $z \in S$.

Some growth restriction is needed: $F(z) = e^{z^2}$ is bounded by 1 on ∂S but unbounded on $\mathbb{R}_+$.

Proof. Subdue the "enemy" e^{z^2} by $e^{z^{3/2}}$: for $\epsilon > 0$ let $F_\epsilon(z) = F(z)e^{-\epsilon z^{3/2}}$ (principal branch, $z^{3/2} = r^{3/2}e^{3i\theta/2}$, $-\pi < \theta < \pi$), holomorphic in S, continuous on $\overline{S}$, with $|e^{-\epsilon z^{3/2}}| = e^{-\epsilon r^{3/2} \cos(3\theta/2)}$. For $|\theta| < \pi/4$, $|3\theta/2| < 3\pi/8 < \pi/2$, so $\cos(3\theta/2)$ is bounded below by a positive constant, and F_ϵ decays rapidly at infinity in $\overline{S}$ and is bounded. Let $M = \sup_{\overline{S}} |F_\epsilon|$; if $F \not\equiv 0$ take w_j with $|F_\epsilon(w_j)| \to M$. As $M \ne 0$ and F_ϵ decays, the w_j accumulate at some $w \in \overline{S}$; by the maximum principle $w \in \partial S$, where $|F_\epsilon| \le 1$. So $|F_\epsilon| \le 1$ on $\overline{S}$; let $\epsilon \to 0$. ∎

Another version of the Paley–Wiener idea characterizes Fourier transforms vanishing for $\xi < 0$:

Theorem 3.5

Let f, $\hat f$ have moderate decrease. Then $\hat f(\xi)=0$ for all $\xi<0$ iff f extends to a continuous bounded function on $\{y\ge 0\}$, holomorphic in $\{y>0\}$.

Proof. If $\hat f=0$ on $\xi<0$, set $f(z)=\int_0^\infty \hat f(\xi)e^{2\pi i\xi z}\,d\xi$ for $y\ge 0$; $|f(z)|\le A\int_0^\infty \frac{d\xi}{1+\xi^2}$, and $f_n(z)=\int_0^n \hat f(\xi)e^{2\pi i\xi z}\,d\xi\to f$ uniformly on the closed half-plane, giving continuity and holomorphy. Conversely, as in Theorem 3.3, let $f_{\epsilon,\delta}(z)=\dfrac{f(z+i\delta)}{(1-i\epsilon z)^2}$, holomorphic near the closed upper half-plane. Shifting the contour up, $\hat f_{\epsilon,\delta}(\xi)=0$ for $\xi<0$; letting $\delta\to 0$ and then $\epsilon\to 0$, $\hat f(\xi)=0$ for $\xi<0$. ∎

Remark

Compare Theorem 7.1: holomorphy in a disc kills Fourier coefficients with $n<0$; holomorphy in the upper half-plane kills $\hat f$ on $\xi<0$.

4 Selected exercises

Motivation. Four exercises from Stein–Shakarchi, Chapter 4 (book numbers kept).

Exercise 4.1 (Uniqueness of the Fourier transform)

Suppose f is continuous and of moderate decrease, and $\hat f(\xi)=0$ for all $\xi\in\mathbb{R}$. Show that $f=0$, as follows. For fixed $t\in\mathbb{R}$ let

$$A(z)=\int_{-\infty}^{t} f(x)e^{-2\pi iz(x-t)}\,dx \qquad (\operatorname{Im} z\ge 0),$$
$$B(z)=-\int_{t}^{\infty} f(x)e^{-2\pi iz(x-t)}\,dx \qquad (\operatorname{Im} z\le 0).$$

(a) Show that $A(\xi)=B(\xi)$ for all $\xi\in\mathbb{R}$.
(b) Show that the function F equal to A in the closed upper half-plane and to B in the lower half-plane is entire and bounded, and that in fact $F=0$.
(c) Deduce that $\int_{-\infty}^{t} f(x)\,dx=0$ for all t, and conclude that $f=0$.

Solution. Write $z=u+iv$, $I=\int_{-\infty}^{\infty}|f(x)|\,dx<\infty$. As $|e^{-2\pi iz(x-t)}|=e^{2\pi v(x-t)}$, the integrand of A is $\le|f(x)|$ for $v\ge 0$, $x\le t$, and that of B for $v\le 0$, $x\ge t$; so $|A|,|B|\le I$ where defined.

(a) For real ξ both are defined, and

$$A(\xi)-B(\xi)=\int_{-\infty}^{\infty} f(x)e^{-2\pi i\xi(x-t)}\,dx=e^{2\pi i\xi t}\hat f(\xi)=0.$$

(b)

Step 1. $A_n(z)=\int_{t-n}^{t} f(x)e^{-2\pi iz(x-t)}\,dx$ is entire (Theorem 5.4 after $x=t-ns$). For $v\ge 0$, $|A-A_n|\le\int_{-\infty}^{t-n}|f|\to 0$, so A is continuous on $\{v\ge 0\}$ and holomorphic in $\{v>0\}$ (Theorem 5.2); likewise B on $\{v\le 0\}$.

Step 2. By (a) they agree on $\mathbb{R}$, so F is entire by the symmetry principle (Theorem 5.5), and constant by Liouville (Corollary 4.5) as $|F|\le I$.

Step 3. For $v > 0$,

$$|F(iv)| = |A(iv)| \le \int_{-\infty}^{t} |f(x)| e^{2\pi v(x-t)}\, dx.$$

The integrand tends to 0 for every $x < t$ as $v \to \infty$ and is dominated by $|f|$, so $F(iv) \to 0$ by dominated convergence. Hence $F \equiv 0$.

(c) In particular $0 = F(0) = A(0) = \int_{-\infty}^{t} f(x)\, dx$ for every $t \in \mathbb{R}$. Since f is continuous, the fundamental theorem of calculus gives $f(t) = \frac{d}{dt}\int_{-\infty}^{t} f(x)\, dx = 0$ for all t. ∎

Exercise 4.7 (Poisson summation for $(\tau+z)^{-k}$)

Fix τ with $\operatorname{Im}\tau > 0$ and an integer $k \ge 2$.
(a) Apply the Poisson summation formula to $f(z) = (\tau+z)^{-k}$ to obtain

$$\sum_{n=-\infty}^{\infty} \frac{1}{(\tau+n)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{m=1}^{\infty} m^{k-1} e^{2\pi i m\tau}.$$

(b) Deduce that $\displaystyle\sum_{n=-\infty}^{\infty} \frac{1}{(\tau+n)^2} = \frac{\pi^2}{\sin^2(\pi\tau)}$ when $\operatorname{Im}\tau > 0$.
(c) Show that the formula in (b) holds for every $\tau \in \mathbb{C} \setminus \mathbb{Z}$.

Solution.

(a)

Step 1. ($f \in \mathfrak{F}$.) Let $t = \operatorname{Im}\tau$, $0 < a < t$. The only singularity $-\tau$ has $\operatorname{Im}(-\tau) = -t$, so f is holomorphic in S_a. For $|y| < a$, $|\tau+z| \ge t - a > 0$ and $|\tau+z| \ge |x| - |\tau|$, so $|\tau+z| \ge c(1+|x|)$ with $c = c(\tau, a) > 0$, and

$$|f(x+iy)| \le \frac{c^{-k}}{(1+|x|)^k} \le \frac{c^{-k}}{(1+|x|)^2} \le \frac{2c^{-k}}{1+x^2}.$$

So $f \in \mathfrak{F}_a$, and Theorem 2.4 gives $\sum_n f(n) = \sum_n \hat{f}(n)$.

Step 2. ($\xi < 0$.) Let $g(z) = (\tau+z)^{-k} e^{-2\pi i z\xi}$; $|e^{-2\pi i z\xi}| \le 1$ for $\operatorname{Im} z \ge 0$. The integral over the boundary of $\{|z| < R,\ \operatorname{Im} z > 0\}$ is 0 (no pole), and the arc contributes at most $\pi R(R - |\tau|)^{-k} \to 0$. So $\hat{f}(\xi) = 0$.

Step 3. ($\xi = 0$.) $\hat{f}(0) = \int_{-\infty}^{\infty} (\tau+x)^{-k}\, dx = \left[\frac{(\tau+x)^{1-k}}{1-k}\right]_{-\infty}^{\infty} = 0$, since $k \ge 2$.

Step 4. ($\xi > 0$.) Close the contour in the lower half-plane, where $|e^{-2\pi i z\xi}| \le 1$; the arc is $o(1)$ and the pole $-\tau$ of order k is encircled clockwise. By Theorem 1.4,

$$\operatorname{res}_{z=-\tau} g = \frac{1}{(k-1)!} \frac{d^{k-1}}{dz^{k-1}} e^{-2\pi i z\xi}\Big|_{z=-\tau} = \frac{(-2\pi i\xi)^{k-1}}{(k-1)!} e^{2\pi i\tau\xi},$$

so

$$\hat{f}(\xi) = -2\pi i \operatorname{res}_{z=-\tau} g = \frac{(-2\pi i)^k}{(k-1)!}\, \xi^{k-1} e^{2\pi i\tau\xi}.$$

Step 5. Summing $\hat{f}(m)$ over $m \ge 1$ (the terms with $m \le 0$ vanish) gives the formula.

(b) Put $k = 2$ and $q = e^{2\pi i\tau}$, so $|q| = e^{-2\pi t} < 1$. Differentiating the geometric series gives $\sum_{m\ge1} m q^m = q/(1-q)^2$, so the right side of (a) equals $-4\pi^2 q/(1-q)^2$. On the other hand,

$$\sin^2(\pi\tau) = \Big(\frac{e^{i\pi\tau} - e^{-i\pi\tau}}{2i}\Big)^2 = -\frac{q - 2 + q^{-1}}{4} = -\frac{(1-q)^2}{4q},$$

so $\pi^2/\sin^2(\pi\tau) = -4\pi^2 q/(1-q)^2$ as well.

(c) For compact $K \subset \mathbb{C}\setminus\mathbb{Z}$, $K \subset \{|z| \le R\}$, and $|n| \ge 2R$, $|\tau+n|^{-2} \le 4/n^2$ on K, so the series is holomorphic on $\mathbb{C}\setminus\mathbb{Z}$ (Theorem 5.2), as is $\pi^2/\sin^2(\pi\tau)$. They agree on the upper half-plane by (b), hence on the connected $\mathbb{C}\setminus\mathbb{Z}$ (Theorem 4.8). ∎

Exercise 4.8 (Power series of a band-limited function)

Let f be continuous and of moderate decrease, with $\hat f$ supported in $[-M, M]$, and let $f(z) = \sum_{n\ge 0} a_n z^n$ be the entire extension given by Theorem 3.3.

(a) Show that $a_n = \dfrac{(2\pi i)^n}{n!}\displaystyle\int_{-M}^{M} \hat f(\xi)\xi^n\, d\xi$, and hence $\limsup_{n\to\infty}(n!|a_n|)^{1/n} \le 2\pi M$.

(b) Conversely, let $f(z) = \sum_{n\ge 0} a_n z^n$ be any power series with $\limsup_n (n!|a_n|)^{1/n} \le 2\pi M$. Show that f is entire and that for every $\epsilon > 0$ there is A_ϵ with $|f(z)| \le A_\epsilon e^{2\pi(M+\epsilon)|z|}$.

Solution.

(a)

Step 1. By the proof of Theorem 3.3, $f(z) = \int_{-M}^{M} \hat f(\xi) e^{2\pi i\xi z}\, d\xi$. Here $\hat f$ is continuous, being the transform of a function of moderate decrease, so $C := \int_{-M}^{M}|\hat f| < \infty$.

Step 2. For fixed z, $e^{2\pi i\xi z} = \sum_n (2\pi i\xi z)^n/n!$ converges uniformly in $\xi \in [-M, M]$ (Weierstrass M-test with $(2\pi M|z|)^n/n!$). So we may integrate term by term:

$$f(z) = \sum_{n=0}^{\infty} \Big(\frac{(2\pi i)^n}{n!} \int_{-M}^{M} \hat f(\xi)\xi^n\, d\xi \Big) z^n.$$

By uniqueness of power series coefficients this is the stated formula for a_n.

Step 3. Hence $n!|a_n| \le (2\pi)^n M^n C$, so $(n!|a_n|)^{1/n} \le 2\pi M C^{1/n} \to 2\pi M$ (if $C = 0$ then $f = 0$).

(b)

Step 1. Fix $\epsilon > 0$ and put $K = 2\pi(M+\epsilon)$. Since $\limsup(n!|a_n|)^{1/n} < K$, there is N with $n!|a_n| \le K^n$ for $n \ge N$. Let $A_\epsilon = \max\big(1, \max_{n<N} n!|a_n|K^{-n}\big)$. Then $|a_n| \le A_\epsilon K^n/n!$ for all n.

Step 2. Hence $\sum|a_n||z|^n \le A_\epsilon \sum (K|z|)^n/n! = A_\epsilon e^{K|z|} < \infty$ for every z. The radius of convergence is infinite, so f is entire, and $|f(z)| \le A_\epsilon e^{2\pi(M+\epsilon)|z|}$. ∎

Exercise 4.9 (Phragmén–Lindelöf in a sector)

Let S be an open sector with vertex 0 and opening angle π/β, where $\beta > 1/2$. Let F be holomorphic in S and continuous on $\overline{S}$, with $|F| \le 1$ on ∂S and $|F(z)| \le Ce^{c|z|^\alpha}$ in S for some $C, c > 0$ and $0 < \alpha < \beta$. Prove that $|F(z)| \le 1$ for all $z \in S$. (For $\beta = 1$ this is the half-plane case; $\beta = 2$ is Theorem 3.4.)

Solution.

Step 1. (Normalization.) A rotation $z \mapsto e^{i\varphi}z$ preserves all hypotheses, so we may assume $S = \{re^{i\theta} : r > 0,\ |\theta| < \pi/(2\beta)\}$. Since $\pi/(2\beta) < \pi$, $\overline{S}\setminus\{0\}$ misses the negative real axis.

Step 2. (Auxiliary factor.) Fix γ with $\alpha < \gamma < \beta$ and let $z^\gamma = r^\gamma e^{i\gamma\theta}$ be the principal branch, holomorphic on S and continuous on $\overline{S}$. For $|\theta| \le \pi/(2\beta)$ we have $|\gamma\theta| \le \gamma\pi/(2\beta) < \pi/2$, so

$$\cos(\gamma\theta) \ge \delta := \cos\Big(\frac{\gamma\pi}{2\beta}\Big) > 0.$$

For $\epsilon > 0$ put $F_\epsilon(z) = F(z)e^{-\epsilon z^\gamma}$. Then $|e^{-\epsilon z^\gamma}| = e^{-\epsilon r^\gamma\cos\gamma\theta} \le e^{-\epsilon\delta r^\gamma} \le 1$ on $\overline{S}$.

Step 3. (Estimates.) On ∂S, $|F_\epsilon| \le |F| \le 1$. In S, $|F_\epsilon(z)| \le C\exp(cr^\alpha - \epsilon\delta r^\gamma) \to 0$ as $r \to \infty$, because $\gamma > \alpha$. Choose R_0 such that $|F_\epsilon| \le 1$ on $\overline{S} \cap \{|z| \ge R_0\}$.

Step 4. (Maximum modulus.) For $R \ge R_0$, $S_R = S \cap \{|z| < R\}$ is bounded with $|F_\epsilon| \le 1$ on its boundary (two segments of ∂S and an arc), so $|F_\epsilon| \le 1$ on S_R (Corollary 4.6), hence on S.

Step 5. For fixed $z \in S$, $|F(z)| \le e^{\epsilon|z^\gamma|} = e^{\epsilon|z|^\gamma}$. Letting $\epsilon \to 0$ gives $|F(z)| \le 1$. ■